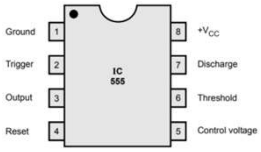
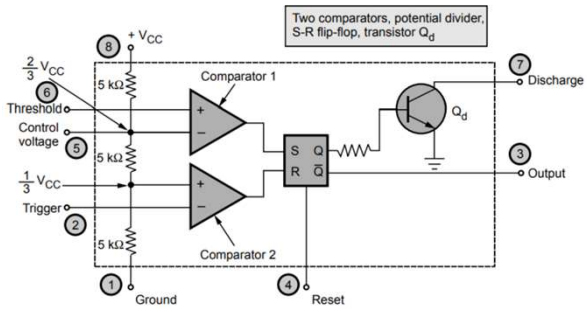
Block Diagram of IC 555

- The 555 timer chip is extremely robust and stable 8-pin device that can be operated either as a very accurate Monostable, Bistable or Astable Multivibrator
- It is used to produce a variety of applications such as one-shot or delay timers, pulse generation, LED and lamp flashers, alarms and tone generation, logic clocks, frequency division, power supplies and converters etc

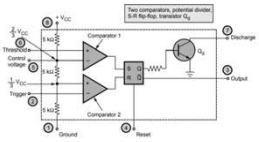


1



2

Pin 1. – **Ground**, The ground pin connects the 555 timer to the negative (0v) supply rail.



• Pin 2. – **Trigger**, The negative input to comparator No 2. A negative pulse on this pin "reset" the internal Flip-flop when the voltage drops below 1/3Vcc causing the output to switch from a "LOW" to a "HIGH" state.

• Pin 3. – **Output**, The output pin can drive any TTL circuit and is capable of sourcing or sinking up to 200mA of current at an output voltage equal to approximately Vcc – 1.5V so small speakers, LEDs or motors can be connected directly to the output.

• Pin 4. – **Reset**, This pin is used to "reset" the internal Flip-flop controlling the state of the output, pin 3. This is an active-low input and is generally connected to a logic "1" level when not used to prevent any unwanted resetting of the output.

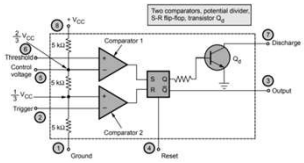
•Pin 5. – **Control Voltage**, This pin can be used to overriding the 2/3Vcc level of the voltage divider network. By applying a voltage to this pin the width of the output signal can be varied independently of the RC timing network. When not used it is connected to ground via a 10nF capacitor to eliminate any noise.

3

• Pin 6. – **Threshold**, The positive input to comparator No 1. This pin is used to reset the Flip-flop when the voltage applied to it exceeds 2/3Vcc causing the output to switch from "HIGH" to "LOW" state.

• Pin 7. – **Discharge**, The discharge pin is connected directly to the Collector of an internal NPN transistor which is used to "discharge" the timing capacitor to ground when the output at pin 3 switches "LOW".

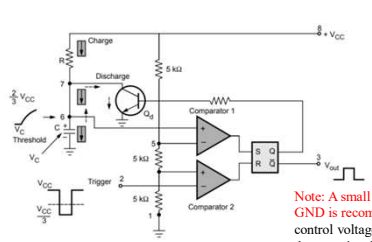
• Pin 8. – **Supply +Vcc**, This is the power supply pin and for general purpose TTL 555 timers is between 4.5V and 15V.

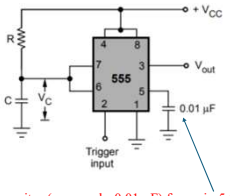


4

Monostable Multivibrator using IC 555

The circuit has only one stable state. When trigger is applied, it produces a pulse at the output and returns back to its stable state. The duration of the pulse depends on the values of R and C. As it has only one stable state, it is called one shot multivibrator.

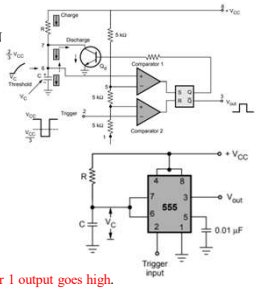




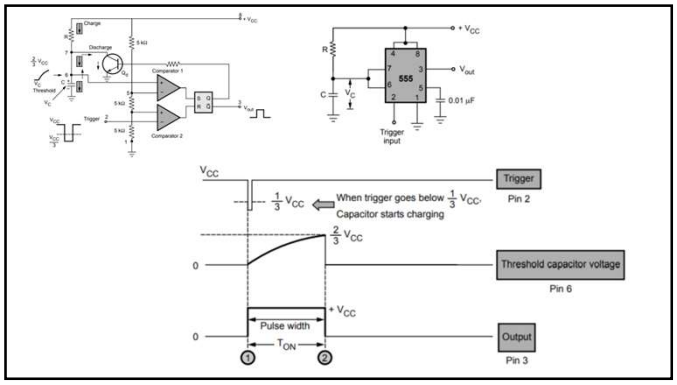
Note: A small capacitor (commonly 0.01 µF) from pin 5 to GND is recommended. Its job is to filter out noise on the control voltage line and stabilize the internal reference levels. If the capacitor is omitted, high-frequency noise could couple into pin 5 and cause erratic triggering or unstable operation.

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1. Assume flip-flop is initially set ➔ Q = 1 and $\bar{Q}$ = 0. ➔ Qd is ON
   - C discharges completely and voltage across it is nearly zero.
   - The output at pin 3 is low.
2. Assume a negative going trigger voltage is less than 1/3 VCC is applied at pin2.
   - When trigger becomes less than 1/3 Vcc, then comparator 2 output goes high.
   - This resets the flip-flop ➔ Q = 0 and $\bar{Q}$ = 1
   - Q = 0 ➔ Qd is OFF.
   - Hence capacitor starts charging through resistance R.
3. Now the voltage a/c C increases exponentially.
   - This voltage is nothing but the threshold voltage at pin 6.
4. When voltage a/c C becomes more than 2/3 Vcc, then comparator 1 output goes high.
   - ➔ Q = 1 and $\bar{Q}$ = 0 ➔ Qd is ON
5. Thus capacitor C quickly discharges through Qd



6



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Derivation of Pulse Width

The voltage across capacitor increases exponentially and is given by,

$$V_C = V\,(1 - e^{-t/CR})$$

If $V_C = 2/3\ V_{CC}$

then $\frac{2}{3} V_{CC} = V_{CC}\,(1 - e^{-t/CR})$

$$\frac{2}{3} - 1 = -e^{-t/CR}$$

$$\frac{1}{3} = e^{-t/CR}$$

$$\therefore \quad -\frac{t}{CR} = -1.0986$$

$$\therefore \quad t = +1.0986\ CR$$

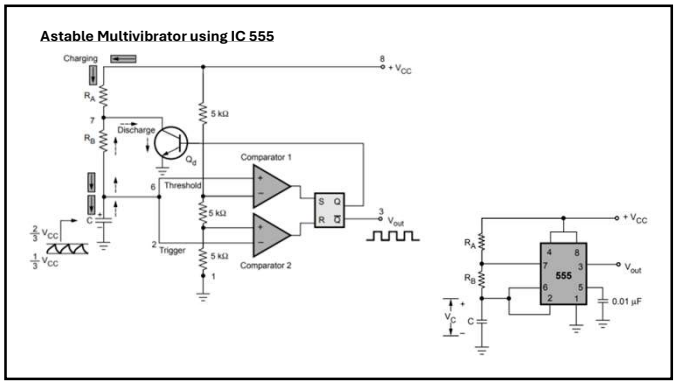
$$\therefore \quad t = 1.1\ CR$$

where C in farads, R in ohms, t in seconds.

Thus, we can say that voltage across capacitor will reach 2/3 VCC in approximately 1.1 times, time constant i.e. 1.1 RC.

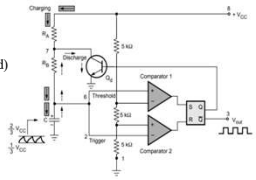
Thus the pulse width denoted as **W is given by, W = 1.1 RC**

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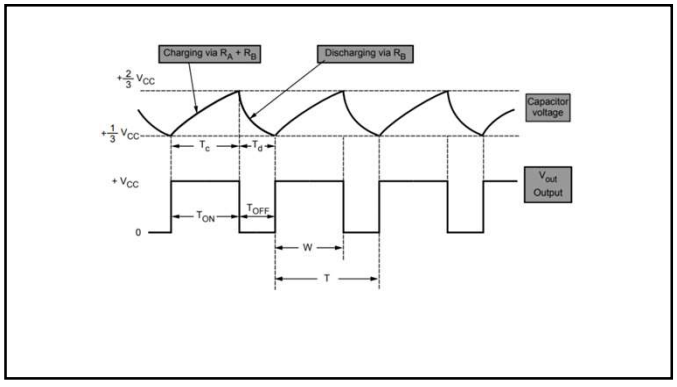


9

- Assume flip-flop is set, ➔ Q =1 and $\bar{Q}$ =0
  - Qd is ON
  - Capacitor starts to discharge. (Assume C was earlier charged)
- Now the capacitor voltage is nothing but the trigger voltage.
  - So while C is discharging, when it becomes less than 1/3VCC, comparator 2 output goes high.
  - This resets the flip-flop hence Q = 0 and $\bar{Q}$ =1 .
- Q =0 ➔ Qd is off.
  - Thus C starts charging through the resistances R_A, R_B and V_{CC}.
  - As total resistance in the charging path is $(R_A + R_B)$, the charging time constant is $(R_A + R_B)$*C.
- Now the capacitor voltage is also a threshold voltage.
- When vtg a/c C = 2/3 VCC, then the comparator 1 output goes high ➔ Q =1 and $\bar{Q}$ =0.
  - Q= 1 ➔ Qd is ON and "C" starts discharging through resistance R_B and transistor Qd.
  - Thus the discharging time constant is R_BC.
  - When vtg a/c C = less than 1/3 VCC, comparator 2 output goes high, resetting the flip-flop.
- This cycle repeats.



10



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Duty Cycle

- Generally the charging time constant is greater than the discharging time constant.
- Hence at the output, the waveform is not symmetric.
- The high output remains for longer period than low output.
- The ratio of high output period and low output period is given by a mathematical parameter called duty cycle.

- W = Time for output is high = T_{ON}
- T = Time of one cycle
- D = Duty cycle = W / T
- %D = W / T × 100 %
- The charging time for the capacitor is given by,
  - Tc = Charging time : 0.693 $(R_A + R_B)$ C
- While the discharge time is given by,
- Td = Discharging time : 0.693 R_B C

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Hence the time for one cycle is,

$$T = T_c + T_d = 0.693 (R_A + R_B) C + 0.693 R_B C$$

$$\therefore \quad \boxed{T = 0.693 (R_A + 2 R_B) C}$$

while $W = T_c = 0.693 (R_A + R_B) C$

$$\therefore \quad \%\ D = \frac{W}{T} \times 100 = \frac{0.693 (R_A + R_B) C}{0.693 (R_A + 2 R_B) C} \times 100$$

$$\therefore \quad \boxed{\%\ D = \frac{(R_A + R_B)}{(R_A + 2 R_B)} \times 100}$$

While the frequency of oscillations is given by, $f = \frac{1}{T} = \frac{1}{0.693 (R_A + 2 R_B) C}$

$$\therefore \quad \boxed{f = \frac{1.44}{(R_A + 2 R_B) C} \text{ Hz}}$$

If R_A is much smaller than R_B, duty cycle approaches to 50% and output waveform approaches to square wave.

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- In the astable multivibrator using 555 timer, $R_A = 2.2$ kΩ, $R_B = 6.8$ kΩ, and C - 0.01 μF. Calculate t_{HIGH}, t_{LOW}, free running frequency and duty cycle.

Solution : $R_A = 2.2$ kΩ, RB = 2.2 kΩ, C = 0.01 μF

$$t_{HIGH} = 0.693 (R_A + R_B) C = 0.693 \times 9 \times 10^3 \times 0.01 \times 10^{-6} = \mathbf{62.37}\ \mu s$$

$$t_{LOW} = 0.693\ R_B C = 0.693 \times 6.8 \times 10^3 \times 0.01 \times 10^{-6} = \mathbf{47.124}\ \mu s$$

$$T = t_{HIGH} + t_{LOW} = \mathbf{109.494}\ \mu s$$

$$\therefore \quad f = \frac{1}{T} = \frac{1}{109.494 \times 10^{-6}} = \mathbf{9.133\ kHz}$$

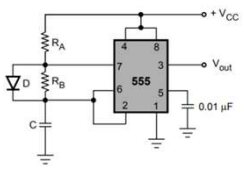
$$\%\ D = \frac{t_{HIGH}}{T} \times 100 = \frac{62.37 \times 10^{-6}}{109.494 \times 10^{-6}} \times 100 = \mathbf{56.962}\ \%$$

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Applications of Astable Multivibrator

Square Wave Generator

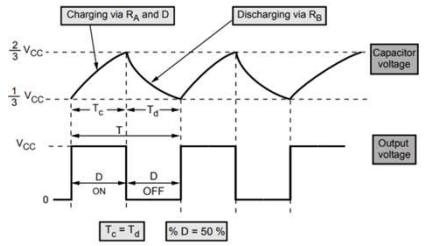
- It can be observed from the expression of duty cycle that in astable operation exact 50 % duty cycle is not possible to achieve. To get exactly 50 % duty cycle i.e. square wave output it is necessary to modify the astable timer circuit.



- In the modified circuit, the capacitor C charges through R_A and diode D and discharges through R_B.
- To obtain square wave (50 % duty cycle) resistance R_B is adjusted such that it is equal to the summation of resistance R_A and the forward resistance of diode D.
- Usually, potentiometer is used for exact adjustments of resistors.

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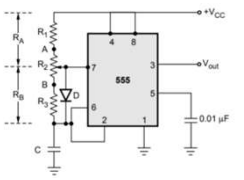
The waveforms of square wave generator



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Astable Multivibrator with a Variable Duty Cycle

- Generally astable mode of IC 555 is used to obtain the duty cycle between 50 % to 100 %. But if duty cycle less than 50 % is required, the circuit can be modified



The charging time for the capacitor is,

$T_c = (R_1 + \text{Part of } R_2)\ 0.693\ C$

The discharging time for the capacitor is,

$T_d = (R_3 + \text{Remaining part of } R_2)\ 0.693\ C$

$T = T_c + T_d = 0.693\ C\ [R_1 + R_2 + R_3]$

$f = 1 / T = 1.44 / (R_1 + R_2 + R_3)\ C$

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Range of duty cycle possible :

Case 1

Let position of R_2 potentiometer is A i.e. part of R_2 in series with R_1 is zero and entirely R_2 gets added to R_3

$$\therefore \quad T_c = R_1\ 0.693\ C$$

$$\text{and} \quad T_d = 0.693 (R_2 + R_3) C$$

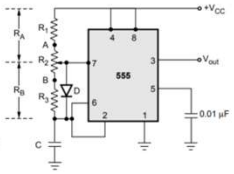
$$\therefore \quad T = 0.693\ C (R_1 + R_2 + R_3)$$

So there is no change in frequency but,

$$\%\ D_{min} = \frac{T_c}{T} \times 100 = \frac{0.693\ R_1 C}{0.693\ C (R_1 + R_2 + R_3)} \times 100$$

$$\therefore \quad \boxed{\%\ D_{min} = \frac{R_1}{R_1 + R_2 + R_3} \times 100}$$

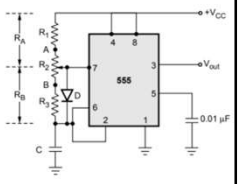
Thus minimum duty cycle, less than 50 % is possible.



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Case 2

For R_2 potentiometer position at B, maximum duty cycle is possible.



$$\therefore \quad T_c = 0.693\,(R_1 + R_2)\,C$$

$$\text{and} \quad T_d = 0.693\,R_3\,C$$

$$\text{and} \quad T = 0.693\,C(R_1 + R_2 + R_3)$$

$$\therefore \quad \%\,D_{max} = \frac{T_c}{T} \times 100 = \frac{0.693\,(R_1 + R_2)\,C}{0.693\,(R_1 + R_2 + R_3)\,C} \times 100$$

$$\therefore \quad \boxed{\%\,D_{max} = \frac{R_1 + R_2}{R_1 + R_2 + R_3} \times 100}$$

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